

Homework 6

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Data: Association Between Bird-Keeping and Risk of Lung Cancer

A 1972-1981 health survey in The Hague, Netherlands, discovered an association between keeping pet birds and increased risk of lung cancer. To investigate birdkeeping as a risk factor, researchers conducted a case-control study of patients in 1985 at four hospitals in The Hague. They identified 49 cases of lung cancer among patients who were registered with a general practice, who were age 65 or younger, and who had resided in the city since 1965. Each patient (case) with cancer was matched with two control subjects (without cancer) by age and sex. Further details can be found in Holst, Kromhout, and Brand (1988).

Age, sex, and smoking history are all known to be associated with lung cancer incidence. Thus, researchers wished to determine after age, sex, socioeconomic status, and smoking have been controlled for, is an additional risk associated with birdkeeping? The data (Ramsey and Schafer 2002) is found in `[birdkeeping.csv(data/birdkeeping.csv)]`.¹

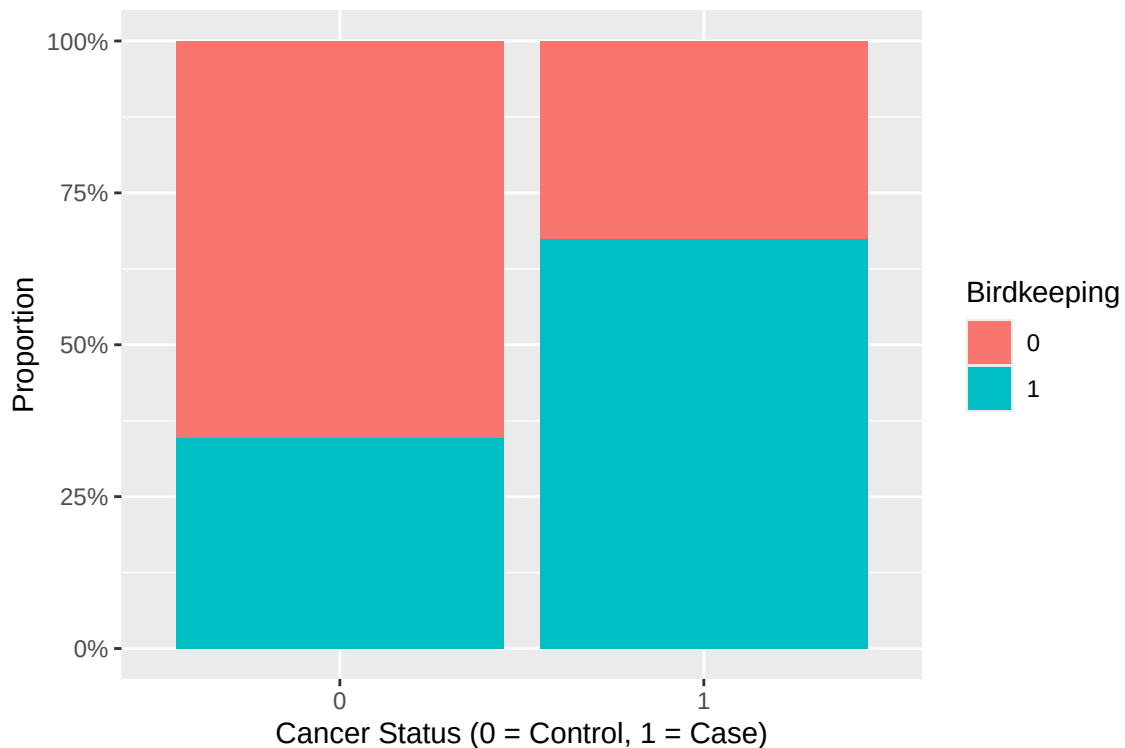
The paper that this exercise is based upon can be found here and please read it before completing the assignment. (<https://www.bmj.com/content/bmj/297/6659/1319.full.pdf>)

Exercise 1

Part a

Create a segmented bar chart and appropriate table of proportions showing the relationship between birdkeeping and cancer diagnosis. Summarize the relationship in 1 - 2 sentences.

¹This problem is adapted from Section 6.8.1, Ex 4.



The chart shows that a much higher proportion of individuals with lung cancer (cases) kept birds compared to those without cancer (controls). This suggests a positive association between birdkeeping and lung cancer, as birdkeeping is more common among cases than controls.

i Part b

Calculate the unadjusted odds ratio of a lung cancer diagnosis comparing birdkeepers to non-birdkeepers. Interpret this odds ratio in context. (Note: an unadjusted odds ratio is found by not controlling for any other variables.)

Fisher's Exact Test for Count Data

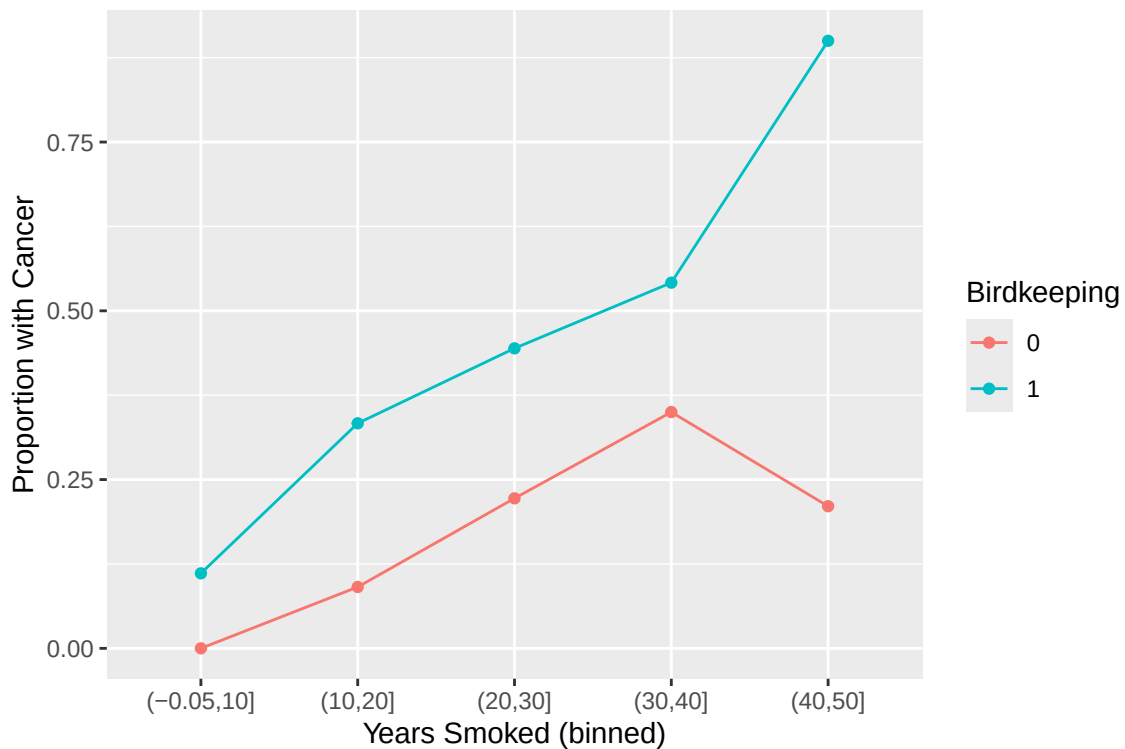
```
data:  tab
p-value = 0.0002188
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
 1.771293 8.630126
sample estimates:
```

odds ratio
3.844044

Individuals who kept birds had about 3.84 times the odds of being diagnosed with lung cancer compared to those who did not keep birds. This suggests a strong positive association between birdkeeping and lung cancer, although this result does not account for other factors like smoking or socioeconomic status.

i Part c

Does there appear to be an interaction between number of years smoked and whether the subject keeps a bird? Demonstrate with an appropriate plot and briefly explain your response.



There appears to be an interaction between years smoked and birdkeeping, as the relationship between smoking and lung cancer differs across the two groups. Specifically, the increase in cancer risk with years smoked is much steeper for birdkeepers than for non-birdkeepers, indicating that smoking has a stronger effect among those who keep birds.

Before answering the next questions, fit logistic regression models in R with cancer as the response and the following sets of explanatory variables:

- model1 = age, yrsmoke, cigsdays, female, highstatus, bird
- model2 = yrsmoke, cigsdays, highstatus, bird
- model3 = yrsmoke, bird
- model4 = yrsmoke, bird, yrsmoke

i Part d

Is there evidence that we can remove age and female from our model? Perform an appropriate test comparing model1 to model2; give a test statistic and p-value, and state a conclusion in context.

```
anova(model2, model1, test = "Chisq")
```

Analysis of Deviance Table

Model 1: cancer ~ yrsmoke + cigsdays + highstatus + bird

Model 2: cancer ~ age + yrsmoke + cigsdays + female + highstatus + bird

	Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
1	142	156.72			
2	140	154.20	2	2.5257	0.2828

A likelihood ratio test comparing model2 to model1 yields a test statistic of $\chi^2=2.53$ with a p-value of 0.2828. Since the p-value is large, we fail to reject the null hypothesis and conclude that there is no strong evidence that age and sex improve the model, so they can be removed from the model.

i Part e

Carefully interpret each of the four model coefficients (including the intercept) in model4 in context.

In model 4, the intercept of -2.9989 represents the log-odds of lung cancer for a person who has never smoked (0 years) and does not keep birds; this corresponds to a very low baseline probability of lung cancer. The coefficient for years smoked (0.0528) indicates that among nonbirdkeepers, each additional year of smoking increases the log-odds of lung cancer by 0.0528. The coefficient for birdkeeping (1.1795) represents the difference in log-odds between birdkeepers and non-birdkeepers when years smoked is zero, meaning that among non-smokers,

birdkeepers have higher odds of lung cancer by a factor of about $e^{1.1795} \approx 3.25$. Finally, the interaction term (0.0093) indicates that the effect of years smoked on lung cancer is slightly stronger for birdkeepers than for non-birdkeepers; specifically, for birdkeepers, each additional year of smoking increases the log-odds by an additional 0.0093 beyond the effect for non-birdkeepers.

i Part f

If you replaced yrsmoke everywhere it appears in model4 with a mean-centered version of yrsmoke, tell what would change among these elements: the 4 coefficients, the 4 p-values for coefficients, and the residual deviance.

The intercept and the bird coefficient will change numerically, because they are now interpreted at the average number of years smoked rather than at 0 years smoked. In contrast, the coefficients for yrsmoke and the interaction term (yrsmoke:bird) will remain the same, since centering does not change slopes. Importantly, none of the p-values for the coefficients will change, because centering is just a linear transformation of a predictor and does not affect statistical significance. The residual deviance will remain exactly the same, since the model fit and predicted values are unchanged.

i Part g

Observe that model3 is a potential final model based on this set of predictor variables. How does the adjusted odds ratio for birdkeeping from model3 compare with the unadjusted odds ratio you found in (b)? Is birdkeeping associated with a significant increase in the odds of developing lung cancer, even after adjusting for other factors?

The adjusted odds ratio for birdkeeping from model3 is approximately 4.37, which is slightly higher than the unadjusted odds ratio of 3.84. This indicates that after adjusting for years smoked, birdkeepers still have substantially higher odds of lung cancer. Additionally, since the birdkeeping coefficient is statistically significant ($p < 0.001$), birdkeeping remains associated with a significant increase in the odds of developing lung cancer, even after accounting for smoking.

i Part h

Discuss the scope of inference in this study. Can we generalize our findings beyond the subjects in this study? Can we conclude that birdkeeping causes increased odds of developing lung cancer? Do you have other concerns with this study design or the analysis you carried out?

The scope of inference is limited because this is an observational case-control study

based on patients from specific hospitals in The Hague, so results may not generalize broadly to other populations. We cannot conclude that birdkeeping causes lung cancer since there was no random assignment and confounding variables may still be present. Although we adjusted for some factors like smoking, unmeasured confounders could still bias the results. Additionally, concerns such as recall bias, selection bias, and a relatively small sample size may affect the reliability of the findings.

Exercise 2

(Ataman and Sariyer 2021) use ordinal logistic regression to predict patient wait and treatment times in an emergency department (ED). The goal is to identify relevant factors that can be used to inform recommendations for reducing wait and treatment times, thus improving the quality of care in the ED.

The data include daily records for ED arrivals in August 2018 at a public hospital in Izmir, Turkey. The response variable is Wait time, a categorical variable with three levels:

- Patients who wait less than 10 minutes
- Patients whose waiting time is in the range of 10-60 minutes
- Patients who wait more than 60 minutes

Part a

Compare and contrast the proportional odds model with the multinomial logistic regression model. Write your response using 3 - 5 sentences. You can find a brief review of the proportional odds model here: <https://library.virginia.edu/data/articles/fitting-and-interpreting-a-proportional-odds-model> and <https://online.stat.psu.edu/stat504/lesson/8/8.4>

The proportional odds model (ordinal logistic regression) is designed for ordered outcomes, like wait times (short, medium, long), and assumes that the relationship between predictors and the odds of being in a higher category is the same across all thresholds (the proportional odds assumption). In contrast, multinomial logistic regression treats the outcome categories as unordered and fits separate models for each category relative to a reference group. As a result, the proportional odds model is more parsimonious and easier to interpret when the ordering is meaningful, while the multinomial model is more flexible but requires estimating more parameters. However, if the proportional odds assumption is violated, the multinomial model may provide a better fit despite being less efficient.

i Part b

Table 5 in the paper contains the output for the wait time and treatment time models. Consider only the model for wait time. Describe the effect of arrival mode (ambulance, walk-in) on the waiting time. Note: walk-in is the baseline in the model. (A link to the paper can be found in the slides).

In the proportional odds model, with walk-in as the baseline, the coefficient for ambulance arrival reflects how arriving by ambulance changes the odds of being in a longer wait time category. A negative coefficient indicates that patients arriving by ambulance have lower odds of experiencing longer wait times compared to walk-in patients. In other words, ambulance patients are more likely to be seen more quickly than walk-ins. This aligns with expectations, as ambulance arrivals are often more urgent and prioritized in emergency departments.

i Part c

Consider output from both the wait time and treatment time models. Use the results from both models to describe the effect of triage level (red = urgent, green = non-urgent) on the wait and treatment times in the ED. Note: red is the baseline level.

In both the wait time and treatment time models, triage level has a significant effect, with red patients as the baseline. Compared to red patients, green patients have higher odds of being in longer wait time categories, meaning they tend to wait longer before being seen. Similarly, for treatment time, green patients are more likely to experience longer treatment durations compared to red patients. Overall, this indicates that urgent patients are prioritized in the emergency department, leading to shorter wait and treatment times relative to non-urgent patients.

Exercise 3

Ibanez and Roussel (2022) conducted an experiment to understand the impact of watching a nature documentary on pro-environmental behavior. The researchers randomly assigned the 113 participants to watch a video about architecture in NYC (control) or a video about Yellowstone National Park (treatment). As part of the experiment, participants played a game in which they had an opportunity to donate to an environmental organization. The data set is available in `nature.csv` in the data folder. We will use the following variables:

- `donation_binary`: 1 - participant donated to environmental organization versus 0 - participant did not donate.
- `Age`: age in years
- `Gender`: Participant's reported gender

- Treatment: Urban (T1)” - the control group versus “Nature (T2)” - the treatment group.
- NEP_high: 1 - score of 4 or higher on the New Ecological Paradigm (NEP) versus 0 - score of less than 4.

See the Introduction and Methods sections of Ibanez and Roussel (2022) for more details about the variables and see the class slides regarding the url for the paper.

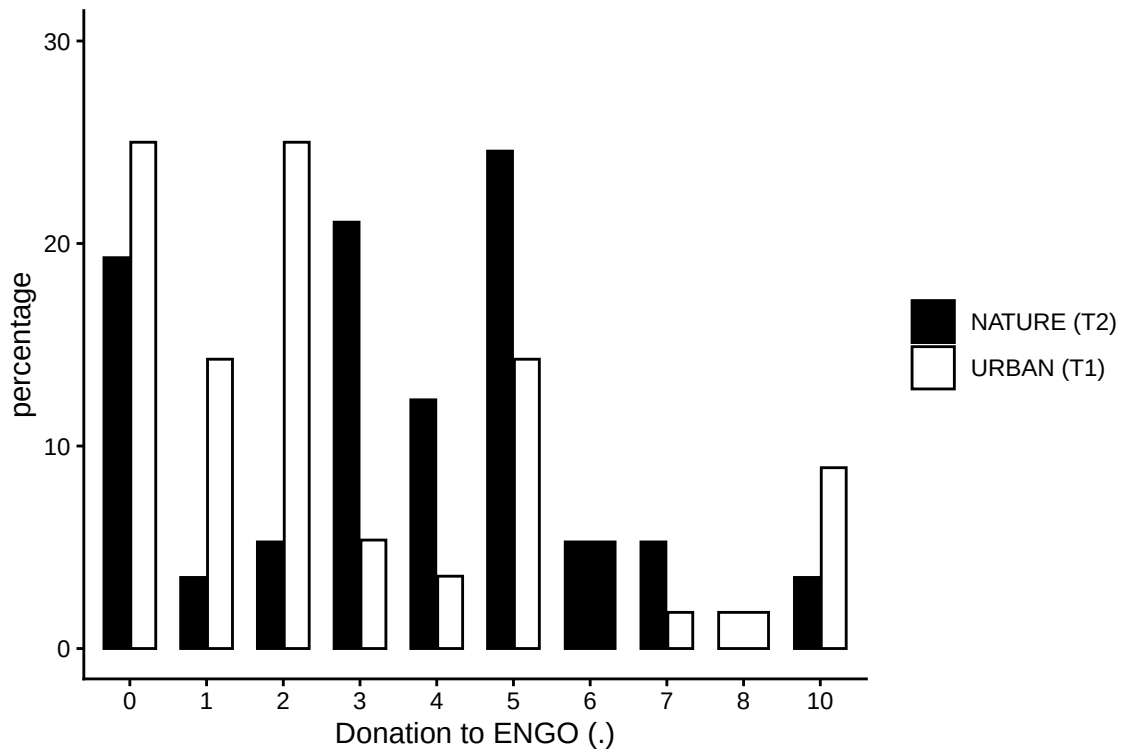
i Part a

Figure 2 on pg. 9 of the article visualizes the relationship between donation amount and treatment. Recreate this visualization using your own code. Use the visualization to describe the relationship between donating and the treatment.

```
library(tidyverse)

plot_dat <- nature %>%
  count(Treatment, `Donation (level)`) %>%
  group_by(Treatment) %>%
  mutate(percent = 100 * n / sum(n)) %>%
  ungroup()

ggplot(plot_dat, aes(x = factor(`Donation (level)`), y = percent, fill = Treatment)) +
  geom_col(
    position = position_dodge(width = 0.7),
    width = 0.65,
    color = "black"
  ) +
  scale_fill_manual(values = c("black", "white")) +
  labs(
    x = "Donation to ENGO (€)",
    y = "percentage",
    fill = NULL
  ) +
  ylim(0, 30) +
  theme_classic()
```

i Part b

Fit a probit regression model using age, gender, treatment, nep_high and the interaction between nep_high and treatment predict the likelihood of donating. (Note: Your model will be similar (but not exactly the same) as the “Likelihood” model in Table 5 on pg. 11.) Display the model.

```
probit_model <- glm(
  donation_binary ~ Age + Gender + Treatment + nep_high + Treatment:nep_high,
  data = nature,
  family = binomial(link = "probit")
)

summary(probit_model)
```

Call:

```
glm(formula = donation_binary ~ Age + Gender + Treatment + nep_high +
  Treatment:nep_high, family = binomial(link = "probit"), data = nature)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.28355	0.99639	0.285	0.7760
Age	0.05974	0.04232	1.412	0.1581
Gender	-0.73733	0.30990	-2.379	0.0173 *
TreatmentURBAN (T1)	-0.19114	0.40559	-0.471	0.6374
nep_high	-0.63860	0.41812	-1.527	0.1267
TreatmentURBAN (T1):nep_high	0.18325	0.56794	0.323	0.7470

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 119.43 on 112 degrees of freedom
Residual deviance: 108.42 on 107 degrees of freedom
AIC: 120.42

Number of Fisher Scoring iterations: 6

i Part c

Describe the effect of watching the documentary on the likelihood of donating.

The effect of watching the documentary is captured by the Treatment (Urban vs Nature) coefficient and its interaction with nep_high. The coefficient for TreatmentURBAN (T1) is negative (-0.191), suggesting that participants in the urban (control) group have a lower latent propensity to donate compared to those in the nature (treatment) group, but this effect is not statistically significant ($p = 0.637$). Additionally, the interaction between treatment and environmental attitude (nep_high) is also not significant ($p = 0.747$), indicating no strong evidence that the effect of the documentary differs by environmental beliefs. Overall, there is no statistically significant evidence from this model that watching the nature documentary increases the likelihood of donating.

i Part d

Based on the model, what is the predicted probability of donating for a 20-year old female in the treatment group with a NEP score of 3?

```
predict(  
  probit_model,  
  newdata = data.frame(  
    Age = 20,  
    Gender = "female",  
    TreatmentURBAN = 1,  
    nep_high = 3  
  )  
)
```

```
    Age = 20,  
    Gender = 1,  
    Treatment = "NATURE (T2)",  
    nep_high = 0  
),  
type = "response"  
)
```

1
0.7706654